BAHIR DAR UNIVERSITY

BAHIR DAR INSTITUTE OF TECHNOLOGY

FACULITY OF COMPUTING

DEPARTMENT OF SOFTWARE

Course: operating system and system programming

Group assignment

Group-12

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1. Introduction

Modern computing is not defined by a single component, but by a sophisticated orchestration of specialized hardware managed by intelligent software. At the foundation of this paradigm is the Operating System (OS), a complex software layer that bridges the gap between raw electronic circuits and high-level user applications.

The OS is tasked with abstracting and managing critical hardware resources, specifically:

- Processors (CPU/SOC): Executing logical instructions and managing computational workloads.
- Memory (RAM): Providing low-latency, volatile workspace for active application data.
- Storage (SSDs): Ensuring reliable, non-volatile persistence of data and programs.
- System Interconnects (Motherboards): Facilitating high-speed, stable communication between all components.

From a system programming standpoint, the OS achieves this control via system calls, kernel-level drivers, and hardware abstraction layers (HAL). This prevents user-space applications from corrupting hardware states while allowing efficient resource multiplexing.

1.1 Integration of Hardware and Software Systems

The close relationship between physical hardware (silicon) and software is what makes modern computing possible.
• Hardware sets the real, physical limits of what a system can do, including its speed, storage, and capacity.
• The operating system (OS) works as a manager, allocating resources like power, memory, and CPU time as needed.
• System programs such as drivers, firmware, and utilities help convert the OS's general commands into specific instructions that hardware devices can understand.

Without this strong integration between hardware and software, features like multitasking, virtualization, and fast data processing would not be possible.

1.2 Hardware Companies Included in This Study

This report evaluates four distinct technology corporations, each dominating a specific stratum of the computing hardware stack:

1. Samsung (Processor Systems)

Samsung designs and manufactures custom ARM-based System-on-Chips (SoCs) under the Exynos brand. Uniquely, Samsung also fabricates these chips, giving them unparalleled vertical integration in semiconductor design and manufacturing.

2. MSI (Computer Systems)

Micro-Star International (MSI) engineers motherboards, specialized PCB layouts, and complete computing systems. They define the physical topologies, power delivery mechanisms, and firmware interfaces that bind components together.

3. Corsair (Memory Systems)

Corsair engineers high-performance DRAM modules. Their products define the speed and latency characteristics of the volatile memory workspace that the OS relies upon for active process execution.

4. Micron Technology (Storage Systems)

Micron is a pioneer in NAND flash memory and high-speed NVMe solid-state drives. Their hardware provides the non-volatile storage foundation required for OS bootstrapping, file systems, and long-term data retention.

1.3 Objective

The core objectives of this analysis are to:

- Deconstruct the architectural contributions of Samsung, MSI, Corsair, and Micron to the computing ecosystem.
- Trace how the Operating System kernel interfaces with and manages these specific hardware architectures.
- Apply system programming concepts (such as DMA, interrupt handling, virtual memory mapping, and I/O scheduling) to real-world hardware scenarios.
- Identify system-level bottlenecks and evaluate how hardware-software co-design mitigates them.

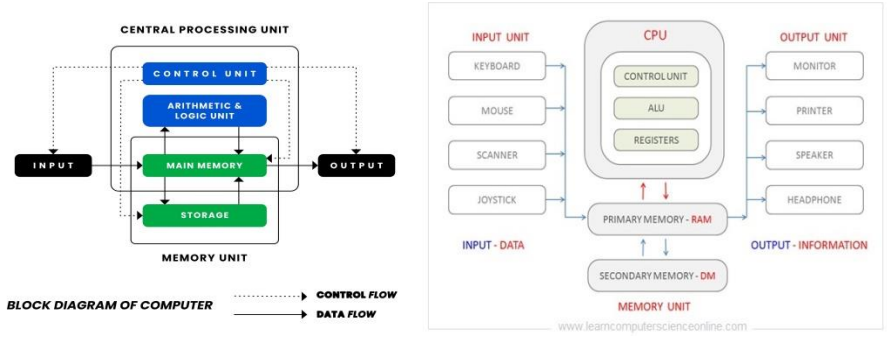


Figure 1: Computer System Block Diagram

2. Processor Systems (Samsung)

Overview

Samsung

Samsung Electronics is a South Korean multinational conglomerate and one of the few companies in the world capable of both designing and manufacturing advanced semiconductor nodes. In the context of processing systems, Samsung's Exynos brand represents a major pillar of ARM-based mobile and embedded computing.

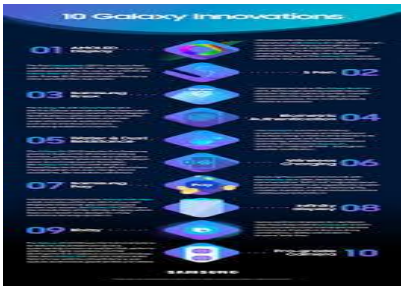
Key Facts

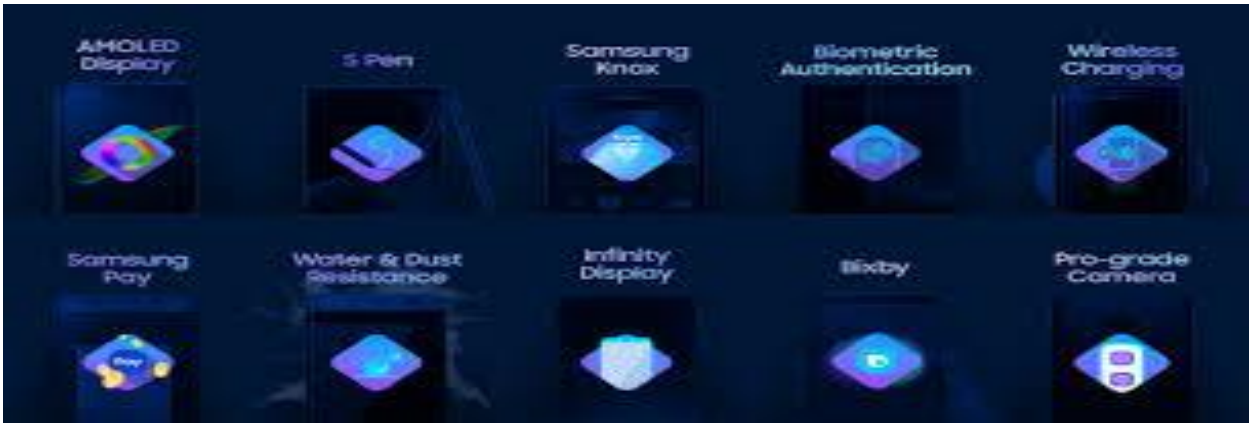
- Founded: 1938 (as a trading company); Semiconductor division launched in 1983.
- Headquarters: Suwon, Gyeonggi-do, South Korea.
- CEO: Jong-Hee Han (Device eXperience division).
- Key Processing Brands: Exynos (Mobile/SoC), IsoCell (Imaging).
- Fabrication Nodes: 3nm GAA (Gate-All-Around), 4nm LPP.

History and Growth

Samsung entered the semiconductor industry by producing DRAM in the early 1980s. By the 1990s, they had expanded into logic chip design. The development of the Exynos line allowed Samsung to reduce reliance on external chipmakers (like Qualcomm) for their flagship Galaxy smartphones. Their unique position as both a fabless designer and a pure-play foundry gives them distinct architectural advantages.

Products and Technologies





Samsung's Exynos processors (such as the Exynos 2400) are highly integrated SoCs featuring custom ARM Cortex CPU cores, AMD RDNA-based Xclipse GPUs, and advanced Neural Processing Units (NPUs). They also pioneer Gate-All-Around (GAA) transistor architecture, which significantly reduces current leakage compared to traditional FinFET designs.

Architecture Explanation

Samsung utilizes the ARMv9 architecture with a heterogeneous multiprocessing (big.LITTLE) approach. Recent iterations employ a 1+2+4+5 tri-cluster configuration:

1 Ultra-Core: For sustained, peak-performance bursts.

2 Performance Cores: For heavy application workloads.

4 Medium Cores: For balanced multi-tasking.

5 Efficiency Cores: For background OS tasks to conserve battery.

OS Interaction

The OS scheduler interacts with Samsung SoCs via the Linux kernel's Heterogeneous Multi-Processing (HMP) framework. The OS must calculate the energy cost of migrating a thread from an efficiency core

to a prime core. The kernel utilizes Demand Response Boost (DRB) to rapidly wake up high-performance cores when interactive UI inputs are detected.

System Programming Perspective

System programmers targeting Samsung hardware write heavily in ARM64 assembly and C. They must optimize for the AMD GPU compute units using OpenCL or Vulkan. Furthermore, programming the NPU requires interfacing with Samsung's OneAPI neural engine, feeding tensor data directly into dedicated AI registers without waking the main CPU.

- Advantages: Unmatched vertical integration (design + fab); leading-edge 3nm GAA power efficiency.
- Limitations: Historically lower peak sustained performance compared to Apple Silicon; complex thermal throttling in thin smartphone chassis.
- Key Insight: Samsung’s control over the fabrication plant allows them to tailor the silicon substrate directly to the OS scheduler’s power constraints.

3. Computer Systems (MSI)

Overview

MSI

Micro-Star International (MSI) is a Taiwanese multinational technology company recognized globally for its gaming hardware, motherboards, laptops, and industrial servers. In the hardware ecosystem, MSI acts as the physical integrator, designing the circuitry that allows disparate components to communicate.

Key Facts

- Founded: 1986, Taipei, Taiwan.
- Founders: Joseph Hsu, Jeans Huang, Frank Lin, Kenny Yu.

- Headquarters: Zhonghe District, New Taipei City, Taiwan.
- CEO: Charles Chiang.
- Key Brands: MSI Gaming (Motherboards/GPUs), Titan, Raider, Stealth (Laptops).

History and Growth

MSI began as a designer and manufacturer of motherboards and add-on cards. Throughout the 2000s, they strategically aligned with the PC gaming boom, developing high-performance motherboard chipsets and eventually expanding into complete gaming laptop and desktop systems. Today, they are a top-tier partner for both Intel and AMD.

Products and Technologies

MSI's core relevance to OS and system programming lies in their motherboards. They design custom PCB traces, multi-layer power delivery systems (VRMs), and proprietary firmware. Their Click BIOS interface is a critical system component that initializes hardware before the OS loads.

Architecture Explanation

MSI motherboard architecture centers on the Chipset (PCH) and the CPU socket topology. They engineer the PCIe lane splitting (e.g., x16 for GPUs, x4 for NVMe SSDs) and DDR memory routing. The physical trace lengths are mathematically calculated to prevent signal degradation (crosstalk) at high frequencies.

OS Interaction

The OS relies on MSI's firmware via the Advanced Configuration and Power Interface (ACPI). When the OS boots, it reads MSI's ACPI tables (DSDT/SSDT) to understand the motherboard's topology—how many USB controllers exist, where the PCIe root complexes are, and the thermal trip points for the VRMs.

System Programming Perspective

System programming for MSI hardware involves writing kernel-mode drivers for proprietary sensors. Software like MSI Center communicates with an onboard Microcontroller Unit (MCU) via the System Management Bus (SMBus). Programmers must write C/C++ code to read I2C registers to fetch fan RPMs and voltages, and write back PWM (Pulse Width Modulation) values to control fan curves.

- Advantages: Exceptional power delivery (VRM) for overclocking; highly robust BIOS interfaces for hardware tuning.
- Limitations: Proprietary software ecosystems can be resource-heavy; complex BIOS settings can lead to system instability if misconfigured by the user.
- Key Insight: MSI represents the physical foundation; without their accurate routing and ACPI reporting, the OS kernel would be blind to the hardware topology.

4. Memory Systems (Corsair)

Overview

Corsair

Corsair is an American computer peripherals and hardware company originally famous for its high-performance DRAM modules. While they have expanded into gaming peripherals, their memory divisions remain a gold standard for PC enthusiasts and system builders.

Key Facts

- Founded: 1994, Fremont, California, USA.
- Headquarters: Milpitas, California, USA.
- Public Listing: NASDAQ: CRSR.
- Key Memory Brands: Vengeance, Dominator Titanium.
- Notable Subsidiaries: Elgato, SCUF Gaming, Origin PC.

History and Growth

Corsair initially manufactured L2 cache modules for PCs. When Intel integrated L2 cache directly onto CPUs, Corsair pivoted to DRAM. They became renowned for screening memory chips (binning) to find those capable of running at higher speeds and lower latencies than standard specifications, creating the enthusiast memory market.

Products and Technologies

Corsair's modern focus is on DDR5 memory. The Dominator Titanium line features custom heat spreaders with RGB, but more importantly, they integrate highly binned memory ICs (often sourced from Samsung, Micron, or SK Hynix) and robust SPD (Serial Presence Detect) chips for extreme overclocking.

Architecture Explanation

Modern Corsair DDR5 modules feature a radically different architecture from DDR4. Each Corsair DIMM now features two independent 32-bit sub-channels (making a 64-bit total bus) and an on-DIMM Power Management IC (PMIC). This means the motherboard no longer regulates RAM voltage; the Corsair stick regulates itself.

OS Interaction

The OS interacts with Corsair RAM through the Memory Management Unit (MMU) and the Page Table structures. The OS allocates physical frames on the Corsair DIMM to virtual pages requested by processes. The OS relies on the timings stored in the module's SPD to correctly program the CPU's memory controller during the boot sequence.

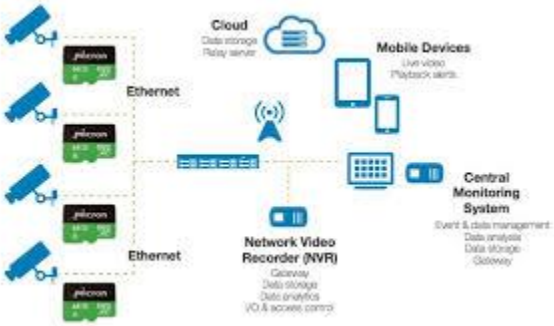
System Programming Perspective

System programmers optimizing for Corsair RAM focus heavily on cache coherence and NUMA (Non-Uniform Memory Access) architectures. They must write code that avoids cache thrashing by aligning data structures to the size of a CPU cache line (typically 64 bytes). If using high-speed Corsair kits, programmers must understand that exceeding the memory controller's tolerance causes system-level Hard Faults (BSODs/Kernel Panics).

- Advantages: On-DIMM PMIC ensures cleaner power delivery to DDR5 chips; stringent binning guarantees stable high-frequency operation.
- Limitations: High-speed DDR5 kits generate significant heat requiring active cooling; premium pricing compared to generic JEDEC memory.
- Key Insight: Corsair's high MT/s ratings are entirely useless if the OS cannot efficiently manage Translation Lookaside Buffer (TLB) misses and page faults.

5. Storage Systems (Micron Technology)

Overview



Micron Technology is an American semiconductor corporation and one of the world's Big Three memory and storage manufacturers. They produce the underlying NAND flash memory and also design complete enterprise and consumer Solid-State Drives under the Crucial brand.

Key Facts

- Founded: 1978, Boise, Idaho, USA.
- Founders: Ward Parkinson, Joe Parkinson, Dennis Wilson, Doug Pitman.
- Headquarters: Boise, Idaho, USA.
- CEO: Sanjay Mehrotra.
- Key Storage Brands: Crucial (Consumer), Micron (Enterprise/Datacenter).

History and Growth

Micron was instrumental in the early development of DRAM. Over the decades, they expanded into NAND flash storage. Micron was the joint developer of the 3D XPoint memory technology (alongside Intel, previously sold as Optane), positioning them at the forefront of storage-class memory innovation.

Products and Technologies

Micron produces 232-layer and 275+ layer 3D NAND flash. Under the Crucial brand, they release drives like the T700, which is a PCIe Gen5 NVMe SSD capable of read speeds exceeding 12,000 MB/s. They also provide flash storage for millions of enterprise servers globally.

Architecture Explanation

Micron's enterprise SSD architecture features a sophisticated on-board storage controller running a Real-Time Operating System (RTOS). They utilize CMOS-under-the-array (CuA) 3D NAND stacking, which places the logic circuitry beneath the memory cells to save space and improve write/amplification management.

OS Interaction

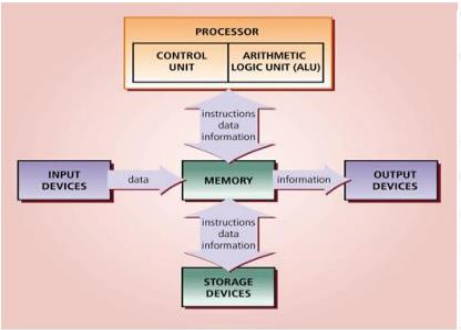
The OS interfaces with Micron SSDs via the NVMe (Non-Volatile Memory Express) protocol over the PCIe bus, managed by the OS Block I/O layer. The OS submits multiple asynchronous read/write requests to multiple submission queues. Crucially, the OS sends TRIM or Deallocate commands to the Micron controller to inform it which logical blocks are deleted, allowing the drive's internal garbage collection to run efficiently.

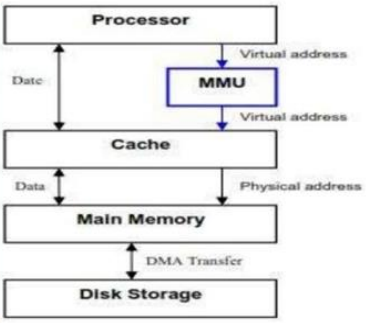
System Programming Perspective

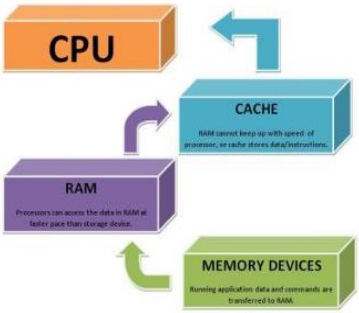
Writing system software for Micron drives requires understanding Direct Memory Access (DMA) and Scatter-Gather Lists (SGLs). System programmers write NVMe driver code that sets up DMA mappings, allowing the Micron SSD to write data directly into Corsair RAM's physical address space without CPU intervention. Modern Linux programmers also utilize io_uring to asynchronously handle massive NVMe queue depths.

- Advantages: Incredible IOPS (Input/Output Operations Per Second); drastically lower latency than traditional SATA SSDs; high reliability in enterprise flavors (PLP - Power Loss Protection).
- Limitations: NVMe Gen5 drives hit severe thermal limits requiring large heatsinks; write speeds can drop significantly when the SLC cache fills up.
- Key Insight: A modern Micron SSD is essentially a specialized, highly parallel computer within a computer, requiring the host OS to intelligently manage I/O queues to prevent bottlenecking.

6. System Integration (Critical Analysis)







A high-performance computing environment is not a collection of independent parts, but a synchronous pipeline. The efficiency of this pipeline depends on how well Samsung, MSI, Corsair, and Micron components communicate under the strict governance of the OS.

6.1 Flow of Data in the System

Micron (Storage) → Corsair (Memory) → Samsung (Processor) → Display Output

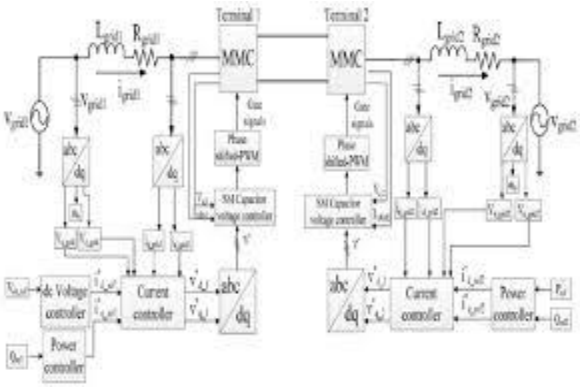
When an application is clicked, the OS sends an I/O request to the Micron SSD. The Micron controller uses DMA to push the application binaries across the MSI motherboard's PCIe lanes directly into the Corsair RAM. The OS then updates the page table, and the Samsung CPU fetches the instructions from Corsair to begin execution.

6.2 System Performance Limitations

If one link in this chain is weak, the entire system slows down:

- MSI Interconnect Limits: If MSI routes the Micron NVMe drive to a slower PCIe Gen3 slot instead of Gen5, storage bandwidth is halved, regardless of the SSD's capability.
- Corsair Latency: If Corsair RAM timings are loose, the Samsung CPU spends more clock cycles waiting for data (memory wall), reducing Instructions Per Cycle (IPC).
- Micron SLC Cache Exhaustion: If a massive file transfer exceeds the Micron drive's dynamic SLC cache, write speeds plummet, stalling OS I/O operations.

6.3 System Power and Temperature Regulation



High-speed data movement generates immense heat. The OS works in tandem with MSI's hardware to prevent thermal throttling:

- The OS reads temperature sensors from the Micron SSD and MSI VRMs.
- If the Samsung CPU package exceeds its thermal design power (TDP), the OS triggers an interrupt, halting non-essential background threads.
- MSI's motherboard firmware dynamically adjusts fan PWM curves to cool the Corsair RAM and Micron SSD heatsinks.

6.4 Role of the Kernel in the System

The kernel acts as the supreme coordinator:

- It schedules threads across Samsung's performance and efficiency cores to balance power.
- It prevents Corsair RAM from overcommitting by enforcing Out-of-Memory (OOM) killer protocols.
- It manages the NVMe submission queues for the Micron drive to ensure real-time OS tasks are not blocked by user file transfers.

6.5 Core Insights

System performance is entirely dictated by the lowest common denominator. The most powerful Samsung SoC cannot process data faster than the MSI motherboard can deliver it, or the Corsair RAM can stage it, or the Micron SSD can supply it. Hardware must be scaled in harmony.

6.6 Summary and Limitations

- Physical signaling limits (PCIe bandwidth, DDR5 trace interference).
- Thermal density constraints in modern form factors.
- OS overhead required to maintain hardware abstraction.
- Incompatibilities arising from proprietary firmware implementations.

7. Comparative Analysis

Feature	Samsung (CPU)	MSI (System)	CORSAIR (RAM)	MICRON (STORAGE)
Primary Role	Instruction Execution	Physical Integration	Volatile Workspace	Non-Volatile Retention
OS Interface	ARM HMP Scheduler	ACPI / UEFI	MMU / Page Tables	Block I/O / NVMe
Programming Focus	Assembly / NPU APIs	SMBus / I2C Drivers	Cache Line Optimization	DMA / io_uring
Critical Metric	IPC / Power Efficiency	Signal Integrity (Trace)	Latency (ns) / MT/s	IOPS / Throughput (MB/s)
Key Vulnerability	Thermal Throttling	Trace Degradation	Page Faults	Write Amplification

8. Future Developments in Computing Systems

8.1 Processing-in-Memory (PIM) Architecture

Future iterations of Corsair and Micron memory will feature embedded compute units directly on the memory die.

- Instead of the OS forcing the Samsung CPU to fetch gigabytes of data from Corsair RAM, simple mathematical operations (like AI matrix multiplication) will occur inside the RAM module.
- This eliminates the memory wall bottleneck and drastically reduces OS scheduling overhead.

8.2 Direct Storage and NVME Optimization

Building on NVMe protocols, Microsoft's DirectStorage allows the Micron SSD to bypass the Samsung CPU entirely for certain workloads (like game asset loading).

- The OS will queue compressed files on the Micron SSD, which will then decompress the data and push it directly to the GPU via the MSI motherboard's PCIe lanes.
- The CPU remains free to handle OS logic and user inputs, fundamentally altering traditional I/O paradigms.

8.3 AI-Driven Operating Systems

Future OS kernels will utilize the AI hardware built into Samsung Exynos chips to become predictive rather than reactive.

- The OS will monitor user habits and proactively page-in applications from Micron storage into Corsair RAM before the user clicks the icon.
- The OS will dynamically adjust the MSI motherboard's power delivery profiles based on predicted workload bursts, optimizing for battery life or performance automatically.

9. Conclusion

This analysis shows that a computing system works as a complete unit, where different hardware components must work together precisely, each with its own specialized role. In this system:

• Samsung provides the core processing power and advanced chip manufacturing.
• MSI supports the system with essential connections, power control, and firmware.
• Corsair delivers fast temporary memory that allows smooth multitasking.
• Micron ensures quick, reliable, and long-term data storage.

However, hardware alone cannot function on its own. It is the Operating System, supported by system programs, that makes these components work together as a complete system. By managing tasks like input/output scheduling, virtual memory, interrupt handling, and temperature control, the OS connects the technical hardware layer with the user's experience.

In the end, the future of computing depends not only on making faster hardware, but also on improving how hardware and software are designed to work together as one unified system.

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